THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET C

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Substances that show different colours in acidic and basic solutions are called: 1

(a) solvents **(b)** catalysts
(c) indicators **(d)** reagents of taste

2. Washing soda is used in soaps because it is: 1

(a) sour **(b)** neutral **(c)** basic **(d)** acidic

3. Which of the following is a neutral substance? 1

(a) Lemon juice **(b)** Lime water **(c)** Distilled water **(d)** Vinegar

4. Which is a safe source for experiments in school? 1

(a) Generator **(b)** Cells
(c) Mains socket **(d)** High-voltage line

5. The two terminals of a cell are: 1

(a) north and south **(b)** positive (+) and negative (−)
(c) red and blue only **(d)** hot and cold

6. Graphite in a pencil is: 1

(a) an acid **(b)** a solvent **(c)** a conductor **(d)** an insulator

7. Which of these is used to make electric heaters' elements? 1

(a) Plastic **(b)** Wood **(c)** Nichrome (alloy) **(d)** Rubber

8. Zinc coating on iron to prevent rust is called: 1

(a) galvanisation **(b)** smelting **(c)** tarnishing **(d)** refining

9. The brown deposit formed on iron in moist air is: 1

(a) salt **(b)** tarnish **(c)** verdigris **(d)** rust

10. When vinegar is added to baking soda we see: 1

(a) bubbles of gas **(b)** a solid ring only
(c) no change **(d)** colour of litmus

11. Which of these is a physical change? 1

(a) Curdling milk **(b)** Dissolving sugar in water
(c) Rusting nails **(d)** Burning wood

12. A change in which new substances are formed is a: 1

(a) chemical change **(b)** temporary change only
(c) physical change **(d)** reversible change only

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 15–25 words each.

13. Name the acids present in vinegar, curd and lemon. 2

14. Name two conductors and two insulators. 2

15. What is lustre? Name two lustrous materials. 2

16. What is crystallisation? Is it a physical or chemical change? 2

17. What is bioluminescence? Give an example. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Explain why we should not use litmus paper made with turmeric for acids. 3

19. What is the role of the switch in a circuit? Give an everyday example. 3

20. Why are bells and musical instruments made of metals? 3

21. Why is food spoilage an undesirable change? How can we prevent it? 3

22. What are the safety rules when working with electricity? 3

OR

Why do electricians use rubber gloves and wooden-handled tools? 3

23. Why is aluminium foil used in packing food? 3

OR

Give three uses of non-metals in daily life. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

24. **Case study:** An electrician repairs a fan with rubber gloves and a plastic-handled screwdriver. 5

(i) Why?
(ii) Which materials are conductors?
(iii) What if the floor is wet?
(iv) Suggest safety steps.

OR

Case study: Riya finds that her torch will not light. She checks the cell, bulb and switch. 5

(i) Draw the circuit components.
(ii) List three possible faults.
(iii) How can she test the bulb?
(iv) Which is a safe energy source for her?

25. (i) Describe five changes around us and classify them. 5
(ii) State whether they are desirable or not.

OR

(i) Define physical and chemical changes. 5
(ii) Give three examples of each.
(iii) How do you identify a chemical change?

END OF QUESTION PAPER — Total: 50 Marks